

Landing Self-Evidencing: Establishing What Actually Ran on a Domestic-Process Mobile NPU

Shusen Yang
Independent Researcher
China
2667199938@qq.com

Abstract—Mobile NPUs are now ubiquitous, but on some vendor platforms the most basic question about a deployment—did the work actually run on the NPU—cannot be answered by measurement. On the platform studied here, NPU utilisation is not exposed to applications, not exposed to the on-device debug shell, and not exposed by any vendor profiling template. Worse, backend placement is silent: a model requested on the NPU may fall back to the CPU with no error code, and a model that does land on the NPU may compute a different answer than its CPU counterpart. We present landing self-evidencing, a measurement protocol that makes backend placement readable from raw logs rather than asserted from configuration, and apply it to a four-stage Chinese licence-plate recognition pipeline on a Kirin 8020 (HarmonyOS 6, MindSpore Lite $\rightarrow$ NNRT). Three findings follow. (i) The protocol is effective: it detected a pure pre-processing defect that was invisible at the output layer, and it correctly attributed that defect to the input rather than to the accelerator. (ii) NPU operator supportability is a model $\times$ toolchain joint property, not a hardware property: the same 51 operator probes that the NNRT path partly rejects pass 51/51 through the CANN toolchain, including ConvTranspose, which cannot even be converted on the NNRT side. (iii) For this pipeline, switching the recogniser’s backend changed 0% of outputs and the detector’s 0.1% over 1,000 real vehicle images—yet a paired McNemar test shows the backend is an accuracy variable (+3.2 pp, $p < 0.0001$), while plate length costs an order of magnitude more (−6.6 pp). We report these results together with their boundaries, including one where a sustained-load experiment failed to reach the thermal regime it was designed to probe.

Index Terms—on-device inference, heterogeneous inference, mobile NPU, HarmonyOS, MindSpore Lite, NNRT, operator delegation, licence plate recognition, measurement methodology

I. Introduction

Deploying a neural network on a mobile NPU is normally presented as a performance question: how much faster is the accelerator than the CPU? On the platform we study, that framing breaks down before it can be applied.

Three properties of the platform obstruct it. First, placement is silent. A session created with an NPU target may be served by the CPU, and nothing in the application-visible result records which happened; a configuration file describing the intended backend is not evidence about the executed backend. Second, acceleration is not guaranteed

by placement. Across a prior characterisation of the same device family, measured speed-ups ranged from $13.11\times$ down to $0.98\times$, with small models losing to the CPU because a fixed per-delegation cost cannot be amortised. Third, and most awkward, a model can land on the NPU and compute the wrong answer without an error code. During an early run, a detector served by the NPU read a plate as SuE05172 where the CPU read SuED5172; both runs returned success.

Under these conditions the ordinary verification instrument is unavailable. On this platform NPU utilisation is not exposed to applications, is not available through the device debug shell, and has no vendor profiling template; the only NPU-related counter available is a temperature, not a load. We therefore treat this as a measurement problem rather than a tuning problem, and ask: what can be honestly established about which hardware executed a given inference, and what follows for the engineering decisions that depend on it?

A. Contributions

- C1 — Landing self-evidencing. A protocol that makes backend placement readable from raw logs: every inference records the requested backend (req), the observed backend (LANDED), and any fallback transition (fallback), and these are read together with a tensor fingerprint and a CPU differential. We describe the protocol, show that it caught a real defect, and—equally important—show the case where it was initially misread, and how that misreading was corrected (§III).
- C2 — Operator coverage is a model $\times$ toolchain joint property. We contrast two toolchains targeting the same silicon. On the NNRT path, 16 of 36 operator probes are rejected when compiled standalone. On the CANN path, the same family of probes passes 51/51. We further characterise the mechanism—operators rejected in isolation are still carried onto the NPU as fusion neighbours of an adjacent convolution—and are explicit that this mechanism is currently supported by correlation, not by a controlled causal experiment (§IV).

- C3 — A layered answer to “should I re-validate after switching backends?”. Over 1,000 real vehicle images, switching the recogniser’s backend changed no output and the detector’s one; but a paired test shows the backend is nonetheless an accuracy variable, and that plate length is a far larger one. We argue the practically correct conclusion is that backend switching requires far less re-validation than is commonly assumed, while licence-plate coverage requires far more (§V).

B. Scope and what this paper does not claim

This work is deliberately narrow. It studies one device, one operating system version, and one application. It reports no utilisation figures, because the platform does not expose any. It makes no cross-vendor claim: the comparison between toolchains in §IV is a comparison of admission boundaries, not of speed. And it reports a sustained-load experiment that did not reach the thermal regime it targeted, because a negative result that is reported is worth more than a positive one that is inferred.

II. Related Work

Mobile on-device evaluation. The most closely related line of work builds automation infrastructure to measure models on real handsets. MELT supports headless benchmarking of language models across Android, iOS and embedded targets, and establishes that on-device execution is memory-bound and thermally limited, with quantisation trading accuracy for viability [1]. That study, and the mobile benchmarking literature generally, targets CPU and GPU execution across Snapdragon/Dimensity/Apple-class silicon. It does not cover domestically produced NPUs, and—crucially for us—it does not have to solve the problem of proving that a specific backend executed, because its targets expose usable counters.

Chinese licence plate recognition. CCPD introduced a large-scale parking-lot dataset with annotations embedded in filenames and an end-to-end detection-plus-recognition baseline; its 2020 extension (CCPD-Green) contains only eight-character new-energy plates [2]. LPRNet showed that segmentation-free recognition is achievable with a lightweight CNN and no RNN, reporting up to 95% on Chinese plates [3]. For distorted and oblique captures, Silva and Jung detect and rectify multiple plates before recognition [4], and Laroca et al. combine a YOLO detector with a recognition stage for real-time operation [5]. Our pipeline is built from public pre-trained weights in this lineage; we train nothing. Our contribution is not a recognition result but a measurement of what happens when such a pipeline is placed on an NPU whose behaviour must first be established.

Operator delegation and fallback. The phenomenon we describe—an operator rejected in isolation yet executing inside a larger graph—is a property of graph partitioning and fusion in the delegation layer rather than of the

arithmetic unit. We are not aware of prior published characterisation of this effect for the specific stack studied here, and we treat our account of it as a hypothesis with supporting observations rather than as an established mechanism (§IV).

III. Platform and Measurement Protocol

A. Platform

All measurements are from a HUAWEI nova 14 Pro (MIA-AL00): Kirin 8020 SoC, HarmonyOS 6.1.0.135, API level 24. Inference runs through MindSpore Lite Kit 2.6.0 (NDK) with the NNRT delegate, reporting the device as `NPU_oalos.boot.hardware.kirin8020_v2_0`. The GPU is a Maleoon 920C accessible only through Vulkan 1.3.275 (nncn). Two additional execution paths are used for control: MindSpore Lite’s CPU backend, and CANN via offline .om compilation.

Because a cross-toolchain comparison is the single easiest thing to get wrong, we state the rule we follow: only same-stack ratios are comparable. NPU numbers come from MindSpore Lite, GPU numbers from nncn-Vulkan, and CPU numbers from two different stacks. Every benchmark table in this paper therefore carries its framework, and cross-stack comparisons are used only as order-of-magnitude sanity checks.

B. Landing self-evidencing

The protocol has three components, which must agree.

(1) Three fields, not one. Each inference emits `req` (what the caller asked for), `LANDED` (what executed), and `fallback` (the transition, if any). A single `LANDED` field is insufficient: `LANDED=CPU` cannot distinguish “the CPU was chosen” from “the NPU was requested and silently fell back”, and those have different remedies.

(2) A tensor fingerprint. We record two L2 norms of the output buffer: one read at the declared dtype, and one read as `fp16`. The intended reading is that a backend which genuinely executed produces a physically-plausible norm, whereas a dtype mismatch produces an implausible one. We initially over-claimed this. Our first formulation asserted that the `fp16`-read norm would be non-zero on the NPU and zero on the CPU. On the device, all three models declare `FP32` outputs, so re-reading them as `fp16` yields garbage on every backend: measured values included NaN on the CPU detector and 10424.0 on the CPU classifier. The correct reading—adopted after that measurement—is that the fingerprint is a mislabel alarm, not a placement criterion. Placement is established by `LANDED`; the fingerprint tells you when the declared dtype is not the actual one.

(3) A CPU differential. For each role we instantiate two sessions, one CPU and one NPU, and compare both latency and output. The differential is what converts “it ran” into “it ran and here is what that cost and changed”.

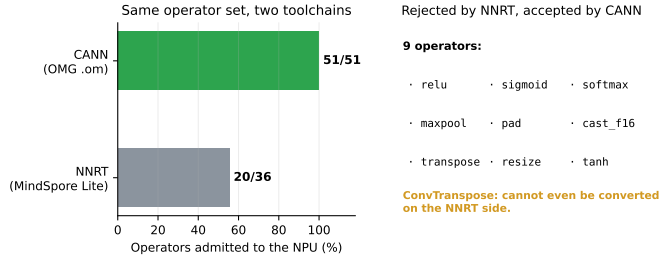


Fig. 1. The same operator family under two toolchains targeting the same silicon. Left: fraction of operator probes admitted to the NPU. Right: the operators rejected by NNRT in isolation but accepted by CANN.

C. Measurement discipline

Latency is reported as the median of 100 runs after 10 warm-up iterations, with ≥ 5 sessions where a session-level claim is made. Thermal level, battery temperature, state of charge and charging state are recorded per round, and comparisons are made only within the same thermal level. Execution order is shuffled with Fisher–Yates where multiple models are involved.

One protocol rule was learned the hard way and is stated here because it generalises beyond this platform. Native code was being compiled with the compiler’s -O0 appended after the build system’s -O3, silently overriding it; there was no warning. The effect was a 1.7–4 $\times$ inflation of measured latency (a colour conversion step measured 25 ms instead of 1.42 ms, and camera throughput 7 fps instead of 20 fps). Every performance number in this paper comes from a build guarded by an automated check that parses the generated build files, extracts the last optimisation flag, and fails if it is -O0.

IV. Operator Coverage Is a Model $\times$ Toolchain Property

A natural way to describe an NPU is by the set of operators it supports. We argue this description is misleading, and that the useful object is the pair (model, toolchain).

On the NNRT path, a set of 36 single-operator probes was compiled standalone and classified by three signals (compilation success, graph build, and execution). Twenty were accepted; sixteen were rejected, including operators as ordinary as ReLU, Sigmoid, Softmax, MaxPool, Pad, Cast, Transpose and Resize. Taken at face value, this says the accelerator cannot execute a rectifier.

It can. We compiled a larger set of 51 operator probes to CANN offline graphs and ran each on the device through three gates (format compatibility, graph construction and build, execution). All 51 passed every gate, including all nine of the operators that the NNRT path rejects. One case is sharper still: ConvTranspose cannot be converted at all on the NNRT side—the converter fails shape inference—yet converts and executes on the CANN side. The same ONNX file fails to produce a model on one path and runs on the other.

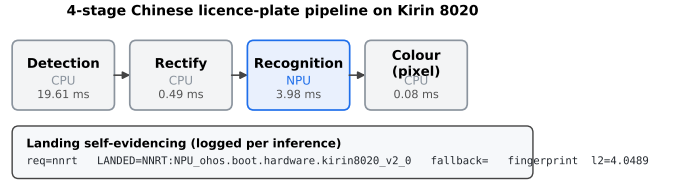


Fig. 2. The four-stage pipeline and the per-inference landing record. Stage timings are from the production configuration on a detected frame (median).

A. Mechanism, and its epistemic status

The observations above are consistent with a specific mechanism. Operators that are rejected in isolation are still carried onto the NPU when they appear as fusion neighbours of an adjacent convolution. The supporting evidence is indirect: residual and MobileNet-family networks contain many rejected operators yet accelerate as a whole by 8–13 $\times$, and partition logs for such models report subgraphs of the form NPU:2, CPU:1 rather than the near-total CPU partitioning that the standalone rejection rate would predict.

We do not claim this mechanism has been causally established. Doing so requires a controlled experiment in which the weights are held fixed and only the isolation of an operator is varied, and measuring the resulting partition and latency. That experiment has not been run. The mechanism is offered as the best current explanation of the correlation, and is flagged as such wherever it is used.

B. What this does and does not license

The comparison licenses a statement about admission: an operator’s absence from one toolchain’s supported list is not evidence that the hardware cannot execute it. It does not license a statement about speed. On the production configuration, MindSpore Lite was in fact faster than CANN for the recognition model (3.99 ms against 5.19 ms). The CANN-side latencies were collected with three repetitions and no warm-up and are therefore used for ordering only, never quoted as measurements.

V. Case Study: A Four-Stage Licence-Plate Pipeline

The application is a Chinese licence-plate recognition pipeline: detection, perspective rectification, sequence recognition, and colour determination. The colour stage is a pixel measurement rather than a classifier, after an investigation established that a pre-trained colour classifier inherited from earlier work was being decoded through a rotated label table—an error that had been recorded as an unresolved disagreement between two components, and which turned out to be an artefact of the decoding rather than a disagreement.

The pipeline is instrumented per stage, so that a single frame yields not one end-to-end number but a budget:

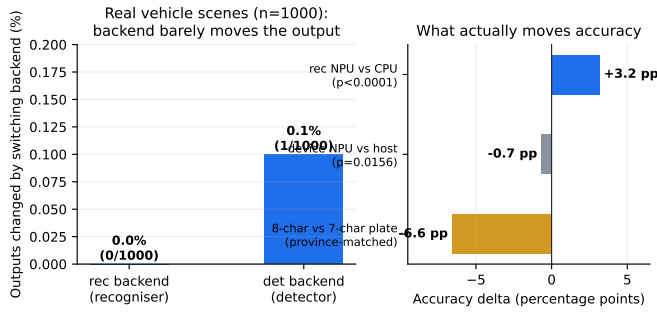


Fig. 3. Left: over 1,000 real vehicle scenes, switching a role’s backend almost never changes the output. Right: the quantities that do move accuracy, including one that moves it in the opposite direction.

on a detected frame in the production configuration, detection accounts for 49% of the frame and recognition for 10%. This decomposition is what makes the rest of the analysis actionable, and it is also what refutes the framing the project began with: on this pipeline, further NPU optimisation has a small ceiling, because the NPU-resident stage is already the cheapest part of the frame.

A. Does switching a backend change the answer?

We ran the same probe over the same 1,000 real vehicle images, changing the backend of exactly one role at a time. Placement was confirmed different by the landing record in each run. Over these images, switching the recogniser’s backend changed zero outputs, and switching the detector’s changed one (0.1%). The single disagreement was a case where both backends were wrong, and wrong differently.

This is the result one wants, and it is also the result that is easiest to over-read. Two further measurements prevent that.

B. Low divergence does not imply absence of bias

Accuracy on the same 1,000 images was 89.9% for the NPU recogniser and 86.7% for the CPU recogniser. A difference of 3.2 pp would be unremarkable if judged against independent confidence intervals, since the two measurements share the images and therefore share the variance attributable to image difficulty. A paired McNemar test, which conditions on the discordant pairs, gives 38 images correct only on the NPU against 6 correct only on the CPU, $p < 0.0001$. Both sessions used the same C++ pre-processing routine and the same model file, so this is a pure backend effect.

The same test applied to the device-versus-host comparison is instructive in the opposite direction. The device NPU recogniser scored 89.9% against 90.6% for the host reference implementation—a 0.7 pp gap, which an earlier analysis of ours had reported as evidence that the port was faithful. The paired test gives 0 images correct only on the device against 7 correct only on the host, $p = 0.0156$. The pairing is strictly nested: $\text{host} \supset \text{NPU} \supset \text{CPU}$. A gap

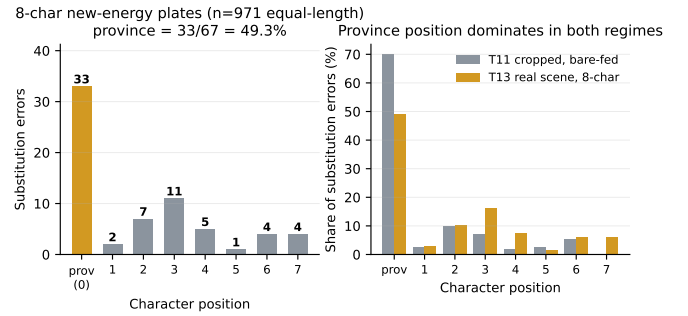


Fig. 4. Substitution errors by character position. The province position dominates across regimes, plate lengths and backends.

of seven images is small and, for engineering purposes, acceptable; but it is systematic, and “faithful” was the wrong word for it.

The general point is that divergence rate and bias are different quantities. A divergence rate of 0.1% that is consistently signed will accumulate into a statistically significant accuracy difference over a thousand images. Reporting only the divergence rate would have hidden this.

C. The dominant variable is plate length, not placement

If backend placement is a second-order effect, what is first-order? Two candidates were measured.

Plate length. On 1,000 eight-character new-energy plates from CCPD-Green, the same pipeline scored 92.5%, against 99.1% on seven-character plates in the comparable real-scene run: a 6.6 pp cost for one additional character, with the recogniser’s backend divergence staying at 0.1%. Because the two datasets differ in province composition, a province-matched control was run: all five provinces with sufficient samples moved in the same direction, and the largest (Anhui, $n = 600$) moved by 6.5 pp, essentially the pooled figure. The cost is attributable to plate length.

Dataset skew. We had hypothesised that the province skew of the inherited benchmark (956/1000 Anhui) was inflating the reported accuracy. On a 26-province balanced set ($n = 3368$) the hypothesis was refuted: accuracy was 90.9% against 90.6%. What skew changes is not how much is wrong but where: on the skewed set, 89% of failures fell on the province position; on the balanced set the province position accounted for a far smaller share and errors moved into the middle of the string. This is a more useful finding than the one we set out to test, and it is the reason the error-position analysis in Fig. 4 is reported per regime.

Across four independent measurements—cropped images fed directly to the recogniser, cropped images through the full pipeline, real vehicle scenes, and eight-character plates—the province position remained the largest single error site (75/77, 5 of 9, and 33/67 = 49.3% respectively), while switching backends left the error pattern essentially unchanged. The province position is a property of the recogniser and the data, not of the accelerator.

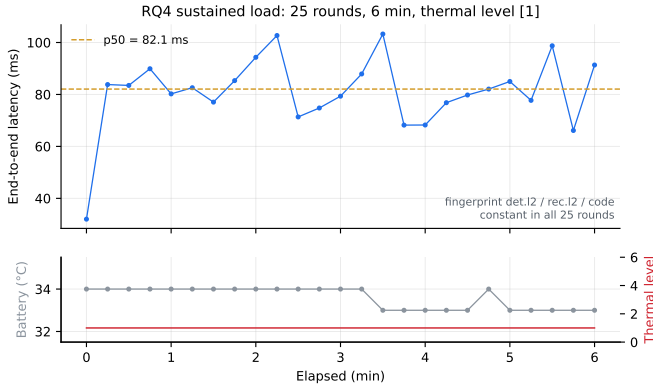


Fig. 5. Sustained load over 80 rounds (23.3 min). Thermal level never leaves level 2 and the tensor fingerprints are constant in every round, yet latency drifts upward. The shaded region is a 213s stall in which the operating system froze the application; it is marked rather than removed, because a reader who did not know about it would read it as a latency spike.

VI. Sustained-Load Behaviour

The fourth question is whether an accelerator’s advantage survives sustained load. On this platform that question is usually framed thermally: the system exposes eight load levels, and the vendor requires that non-essential background work stop from level three upward. A previous characterisation of the same device family reported a 15–20% NPU latency penalty when the thermal level moves from 2 to 3, which makes the crossing a natural quantity to measure.

We ran the production pipeline for 80 rounds at 15s intervals, 23.3 minutes in total, recording thermal level, battery temperature, state of charge, charging state and the landing fingerprints on every round.

A. The thermal regime was not reached

The thermal level was constant at 2 for all 80 rounds; battery temperature stayed within 1 °C. The 2→3 crossing was never reached, so the 15–20% penalty rule is neither confirmed nor refuted here. An earlier 25-round run of the same probe had the same outcome at thermal level 1 over 6 minutes; extending the run by a factor of four did not change it. We report this as a negative result rather than extrapolating.

B. Latency degrades without a thermal crossing

The reason the negative result matters is that latency nevertheless drifted upward, monotonically, over the run:

Two things follow. First, a stable thermal level does not imply stable performance; a measurement protocol that records only thermal state and assumes the rest is stationary will be wrong by roughly a quarter over twenty minutes. Second, the degradation is not uniform across the pipeline: the CPU-resident stages degrade by 29.3% against 19.2% for the NPU-resident stage. Since the production configuration places detection on the CPU,

TABLE I

Latency drift across the sustained run, excluding the frozen interval. Thermal level 2 and battery temperature 35 °C throughout.

	rounds 0–17	rounds 36–53	change
End to end (mean)	64.44 ms	81.76 ms	+26.9%
CPU-resident stages	49.47 ms	63.94 ms	+29.3%
NPU-resident stage	14.92 ms	17.78 ms	+19.2%

this is consistent with the frame-budget analysis in §V and reinforces the same conclusion from a different direction: on this pipeline, the accelerator is not where the variability lives.

We do not attribute the drift to a mechanism. The device does not expose clock or thermal-zone readings to applications, so frequency scaling, scheduler behaviour and background interference cannot be separated here.

C. Two protocol defects found while running this

The operating system freezes long-running foreground applications. Between rounds 54 and 55 the interval was 213s rather than 15s. The device log shows the application being re-activated by the window manager at the end of that interval, implying it had been backgrounded and frozen. This is the same failure mode we had previously observed with screenshot capture. The remedy is to override the screen-off timeout before starting the run (power-shell timeout -o, restored afterwards); a rule of “do not take screenshots” is insufficient, because the trigger here was the screen turning off, not a screenshot. We mark the interval in Fig. 5 rather than removing it, and we exclude it from Table I.

Inter-run variance is not a stable quantity at this sample size. We initially intended to report whether the NPU-resident or CPU-resident stages exhibit more round-to-round variance, following the observation in prior work that CPU-side variance is the larger. Across three measurements of the same quantity we obtained three different answers: CPU-dominant, approximately equal, and NPU-dominant. Segment-wise, the NPU-side coefficient of variation ranges from 8.8% to 48.2% within a single run. We therefore report no comparison of this kind. Answering it requires repeated measurements within a round rather than one measurement per round, which the probe’s design does not provide. What can be stated is simply that round-to-round variance is large (coefficient of variation around 27%), which is why single-point latency figures are not usable as results on this platform.

VII. Discussion and Threats to Validity

Isolated benchmarks are not product-path costs. A discrepancy we did not expect, and which changed how we read every other latency number, is that the same model on the same backend with the same thread count costs roughly twice as much inside the application as in an

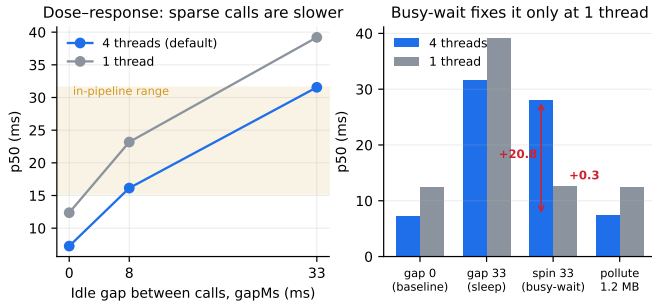


Fig. 6. Why an isolated benchmark is not the product-path cost. Left: raising the idle gap between calls from 0 to 33 ms raises the median from 7.26 ms to 31.56 ms, and the upper end of that range coincides with the in-pipeline measurements. Right: a busy-wait of the same duration does not recover the loss at four threads (+20.8 ms) but almost fully recovers it at one (+0.29 ms), separating frequency scaling from thread-pool wake-up.

isolated probe: 19.5–31.6 ms in the pipeline against 7.4 ms in a tight loop (Fig. 6). A three-step experiment isolated two additive mechanisms. A dose-response sweep showed that increasing the idle gap between calls raises the median from 7.26 ms (no gap) to 16.14 ms (8 ms) to 31.56 ms (33 ms)—and the 33 ms point lands inside the observed in-pipeline range. Replacing the sleep with a busy-wait of the same duration did not restore performance at four threads, which rules out frequency scaling as the sole cause. Repeating at one thread, where there is no worker pool to park, restored it almost completely (a residual of +0.29 ms against +20.8 ms at four threads). The two mechanisms are therefore frequency scaling, which is not controllable from the application, and thread-pool wake-up latency, which is controllable only by keeping threads busy—a remedy that would violate the platform’s rule against sustained background computation. The practical consequence is that an isolated microbenchmark must not be quoted as the cost of a stage in a deployed pipeline.

Throughput claims require knowing what limits the measurement. In the camera path, arrival rate and completion rate are equal and no frames are dropped, which is naturally read as “the pipeline keeps up with the camera”. It is not a valid inference: when the pipeline is the slower party, the receiver buffer fills and the camera stops delivering, so the discarded frames are never counted as dropped. The distinction requires a control configuration that acquires frames without inference; under that control the camera supplied 49.98 fps, while the same configuration with inference reached 36.2 fps (production) and 43.6 fps (all-NPU). The pipeline was the limit, and the arrival rate was reporting our own consumption.

Cross-stack comparability. As stated in §III, only same-stack ratios are comparable. We report a $4.46\times$ speed-up for the recogniser (same stack, same pre-processing, same model file, 16.78 ms against 3.76 ms); we do not report cross-stack speed-ups as results.

What the fingerprint does not prove. The tensor fin-

gerprint establishes that a backend executed. It does not establish that the computation was correct. Correctness is established only against human-annotated ground truth. We note this because our own project contains a case where the fingerprint was initially read as evidence about the accelerator when it was in fact evidence about our own input tensor: a padding region in a pre-processing routine was not being zeroed, so a preceding run through a differently-laid-out buffer left residue in it. The output-layer symptom was a plausible-looking plate string missing one character. The fingerprint faithfully recorded that something changed; only aligning the session timeline showed that the something was on our side.

Scale and generality. All results are from one device, one OS version, and one application. The 1,000-image sets are single-source, and the eight-character set is 60% Anhui, a skew that cannot be removed by sampling because the source data is itself skewed. The cross-SKU question—whether these findings hold across the 9010/9020/9030 parts—was deliberately scoped out because it depends on retail demonstration devices whose availability, thermal state and charging state we cannot control.

VIII. Conclusion and Future Work

We set out to answer a question that this platform makes surprisingly hard: did the accelerator actually execute the work? The answer we can support is that placement can be established from raw logs, provided the request, the outcome and any fallback are recorded separately and corroborated by a differential against the CPU. Doing so caught a real defect, and—at least as usefully—it caught us over-claiming what the fingerprint proves.

Applying the protocol produced a result that is more useful than a speed-up table. For this pipeline, backend placement is nearly inert: it changes almost no outputs, and where it does bias accuracy, it does so by a few percentage points. Plate type and plate length move accuracy by considerably more. The engineering conclusion is that effort spent re-validating outputs after a backend switch is largely misdirected, whereas effort spent on coverage—which plate types, which provinces—is not.

Three directions remain open, and we prefer to name them than to imply they are done. The fusion-carry-over mechanism requires a controlled experiment with fixed weights and varied operator isolation before it can be stated as causal. The sustained-load question—whether the NPU’s advantage survives a thermal soak—requires a longer run than the one we report, and is not answered here. And the cross-SKU scaling question requires hardware we cannot control.

References

- [1] S. Laskaridis, K. Katevas, L. Minto, and H. Haddadi, “MELTing Point: Mobile Evaluation of Language Transformers,” in Proc. 30th Annu. Int. Conf. Mobile Computing and Networking (MobiCom ’24), 2024, pp. 890–907. doi: 10.1145/3636534.3690668.

- [2] Z. Xu, W. Yang, A. Meng, N. Lu, and H. Huang, "Towards End-to-End License Plate Detection and Recognition: A Large Dataset and Baseline," in Proc. European Conf. Computer Vision (ECCV), 2018, pp. 255–271.
- [3] S. Zherzdev and A. Gruzdev, "LPRNet: License Plate Recognition via Deep Neural Networks," arXiv:1806.10447, 2018.
- [4] S. M. Silva and C. R. Jung, "License Plate Detection and Recognition in Unconstrained Scenarios," in Proc. European Conf. Computer Vision (ECCV), 2018, pp. 580–596. doi: 10.1007/978-3-030-01258-8_36.
- [5] R. Laroca, E. Severo, L. A. Zanlorensi, L. S. Oliveira, G. R. Gonçalves, W. R. Schwartz, and D. Menotti, "A Robust Real-Time Automatic License Plate Recognition Based on the YOLO Detector," in Proc. Int. Joint Conf. Neural Networks (IJCNN), 2018.